

STATS 101C Final Project Report: Predicting Obesity

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1. Abstract

Our project focused on exploring a dataset with lifestyle, demographic, and health-related attributes to leverage predictive modeling techniques to predict obesity. The training dataset includes 32,014 entries with variables such as age, gender, dietary habits, physical activity levels, biometric measurements, and obesity status. The testing dataset includes 10672 entries with all the same variables as the training dataset, except obesity status. Our report will detail how we built our classification model spanning from EDA to data cleaning to variable and model selection to a discussion of our results and limitations.

After investigating machine learning models such as logistic regression, decision trees, and random forests we evaluated the performance of these models and concluded that the random forest model was best. Ultimately we ended with a kaggle score of 0.997 and rank 15.

2. Introduction

Obesity is a chronic disease that happens when an individual has an abnormally excessive amount of body fat. This disease poses significant risks for other life threatening illnesses such as hypertension, cardiovascular diseases, diabetes, cancer, and various musculoskeletal disorders. It is a major global health issue, especially in the US, with obesity rates steadily increasing over the past few decades. Early detection and prevention of obesity can mitigate these health risks and improve overall public health. However, in order to identify individuals at risk of obesity, we must analyze the interactions between demographic, lifestyle, and health-related factors.

The Kaggle competition provided testing and training dataset with diverse features, including age, dietary habits, physical activity levels, and family history, to help develop predictive models for obesity. The competition tasked participants with using these features to build predictive models that accurately classify an individual's obesity status. This report details the data exploration process, key findings, and the performance of various predictive models to identify the most effective solution. By contributing to the understanding of obesity prediction, this project highlights the potential for data-driven approaches in addressing critical health issues.

3. Data Analysis

3.1 EDA

We started our EDA of the training dataset by first inspecting the characteristics of the data frame with the functions *summary(data)* and *str(data)* to get an understanding of the types of variables we were working with. Next we visualized the target variable's (Obesity Status') distribution.

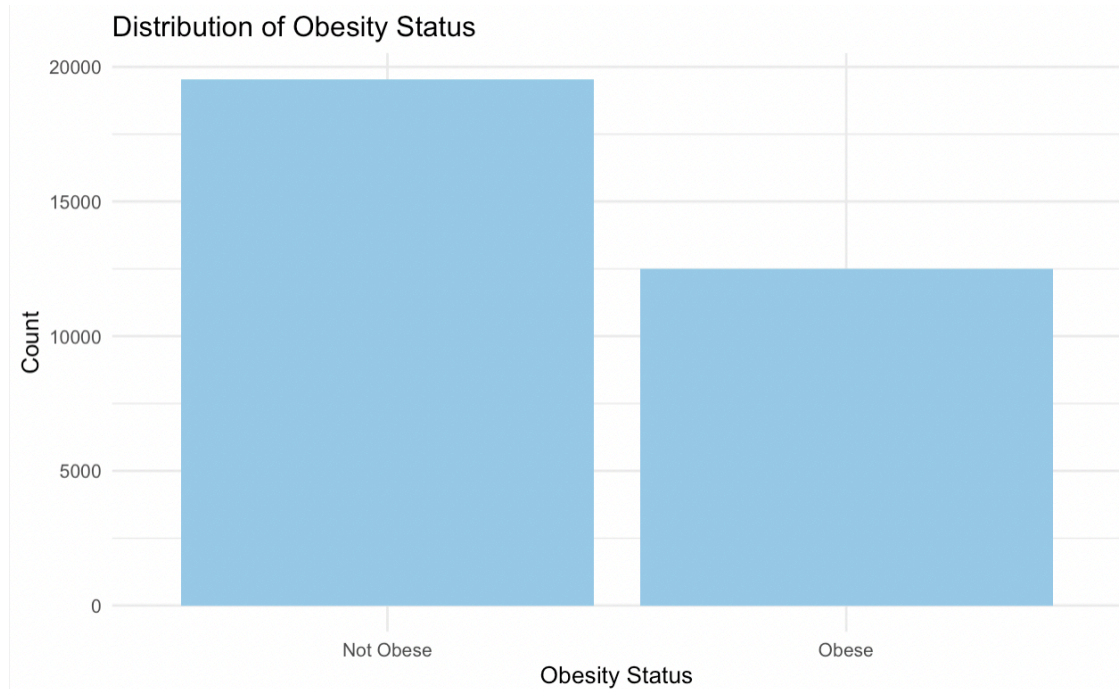


Figure 1: Distribution of Obesity Status: The graph above shows that the majority of individuals in the dataset are classified as "Not Obese". The "Not Obese" group has nearly double the count compared to the "Obese" group. This imbalance suggests the dataset is skewed, which may impact the predictive modeling for obesity status.

Next we looked more specifically at numerical predictors. Since the response variable is binary we created boxplots of the distributions of numerical variables for the different obesity statuses:

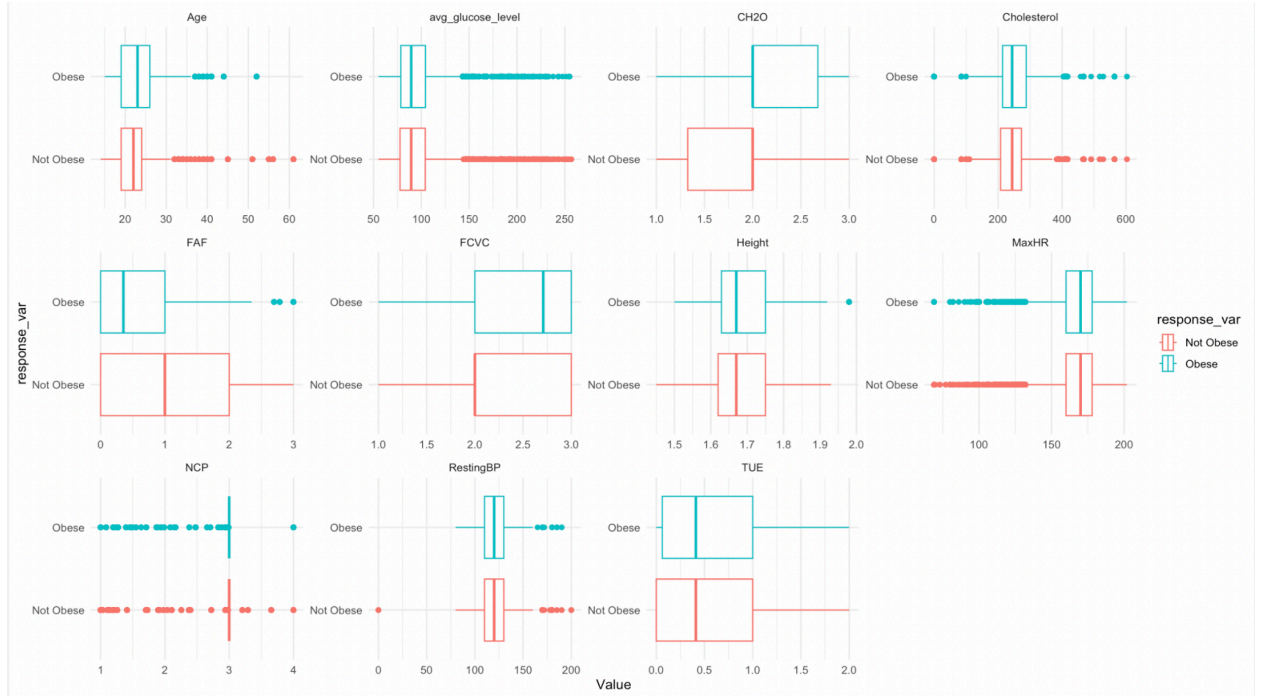


Figure 2: Distribution of Numerical Variables for Each Obesity Status : The boxplots show distinct differences in numeric features between "Obese" and "Not Obese" groups. Some variables, like avg_glucose_level, Cholesterol, and RestingBP, showed higher medians and greater variability for the "Obese" group compared to the "Not Obese" group. Other features such as Height and TUE had lower medians for the "Obese" group. These patterns suggest strong relationships between certain numeric variables and obesity status, which could be useful for predictive modeling and that we will further explore in our variable selection section.

We then conducted EDA on the categorical variables. To start we looked at the different levels for each variable, which is displayed in the table below:

Variable	Number of Levels
CAEC	4
CALC	4
FastingBS	2
Heart Disease	2
family_history_overweight	2
Gender	2
FAVC	2

SMOKE	2
SCC	2
MTRANS	5
Race	4
RestingECG	3
ExerciseAngina	2
hypertension	2
ever_married	2
work_type	5
Residence_type	2
stroke	2
Obesity Status	2

Table 1: Number of Levels for Each Categorical Variable

We then explored the relationship between each categorical variable and obesity status by creating bar graphs that showed the frequency of each level of each categorical variable for each obesity status. We interpreted each one and bear in mind our findings for the variable selection process. Three examples of these bar graphs are shown below:

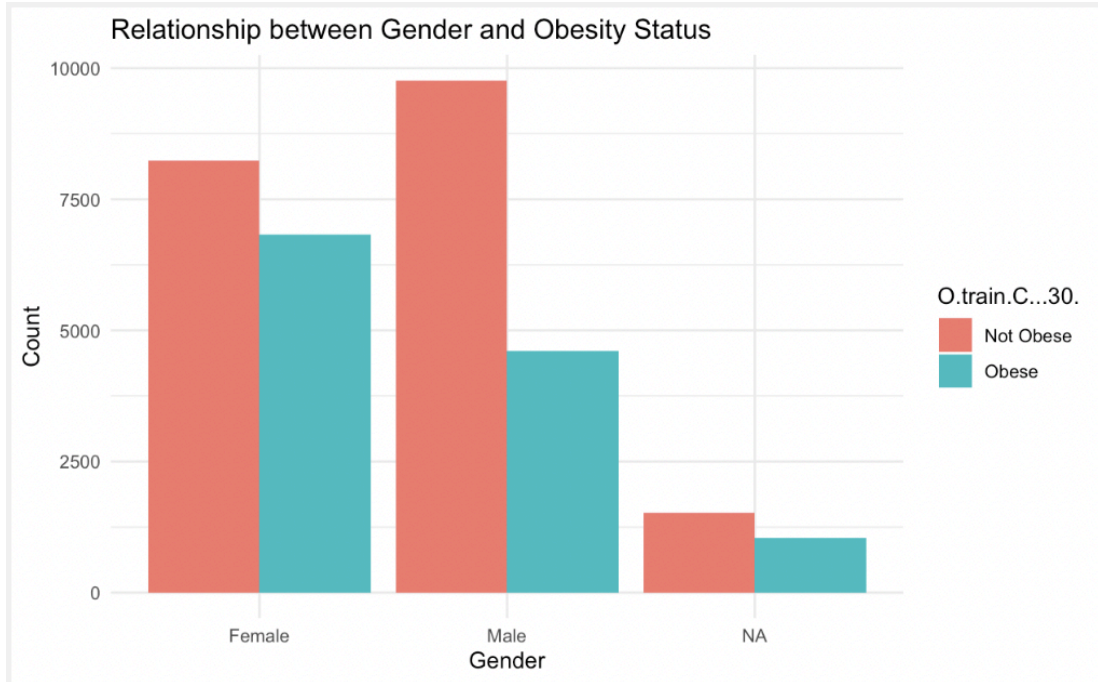


Figure 3: Relationship Between Gender and Obesity Status : The graph shows the distribution of obesity status across gender categories: Female, Male, and NA (missing data). More males are categorized as "Not Obese" compared to "Obese," with a noticeable gap between the two groups. For females, the difference between "Not Obese" and "Obese" is smaller but still significant. Gender appears to play a role in obesity status, with distinct patterns across the groups.

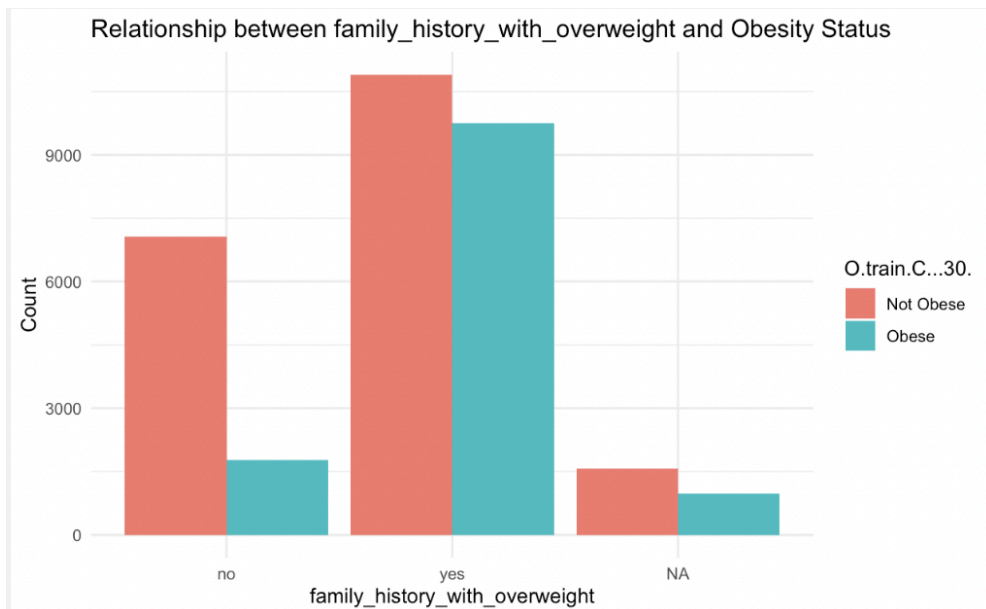


Figure 4: Relationship Between family_history_with_overweight and Obesity Status:

The graph shows the relationship between family history of being overweight and obesity status. Individuals with a family history of being overweight are more likely to be categorized as "Obese" than "Not Obese." On the other hand, individuals without a family history of being overweight are overwhelmingly categorized as "Not Obese." The NA group, representing missing data, has a relatively balanced distribution between "Obese" and "Not Obese," though the counts are much lower. Family history of being overweight strongly correlates with obesity status in this dataset.

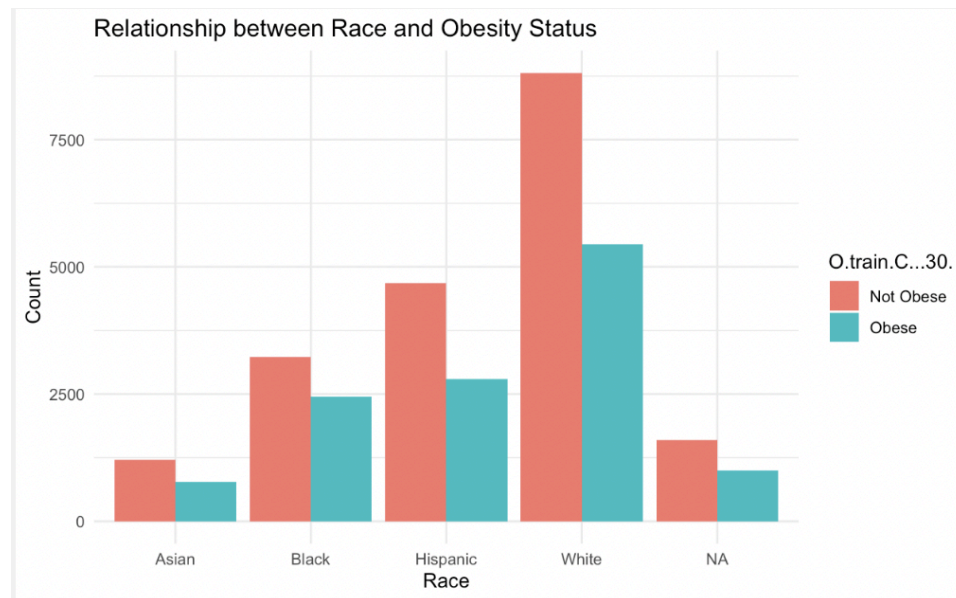


Figure 5: Relationship Between Race and Obesity Status : The graph above shows how obesity status varies across different races. White individuals have the highest count in the "Not Obese" category, with a significant gap compared to the "Obese" category. Hispanic individuals show a smaller but noticeable difference, with more categorized as "Not Obese" than "Obese." Black individuals have a relatively balanced distribution between "Not Obese" and "Obese." Asian individuals have much lower counts in the dataset, but also display a similar pattern where "Not Obese" outnumbers "Obese."

3.2 Data Cleaning

Next, we preprocessed the data, specifically we dealt with missing values because the machine learning algorithms we investigated don't support missing values. Looking more closely at the training and testing data set, we found that all predictors had a small proportion of missing values. The percentage of missing values in each column ranged from 7.496% to 8.602%. Thus,

we decided that we did not need to eliminate any predictors immediately and could impute the missing values as stated previously. We imputed numerical values with the median and categorical values with the mode as these metrics are representative of the overall distribution of each predictor.

3.3 Variable Selection

Lastly, we conducted basic feature importance analysis to get an understanding of what variables we should select for our model. It is important to note that we did not strictly use the results of this *initial* variable selection in each of our models, rather it served as a basis for the variables we selected for each method we investigated.

For the variable selection process, we used AIC and BIC to evaluate the contribution of each feature to the predictive performance of our model. Both metrics serve as criteria for model selection, balancing goodness of fit with model complexity. AIC focuses on minimizing information loss, favoring models that explain the data well without overfitting, while BIC incorporates a stronger penalty for complexity, favoring simpler models when datasets are large. Our approach systematically assessed each variable in the dataset by fitting individual logistic regression models and calculating their AIC and BIC values. The process began by iterating through all variables, excluding the target variable, and fitting logistic regression models for each predictor individually.

After fitting each model, AIC and BIC values were extracted and recorded in the table below. We sorted results by AIC to identify variables that contributed the most to explaining the variance in the target variable. Features with the lowest AIC and BIC scores were considered strong candidates for inclusion in the final model, as they struck an optimal balance between fit and simplicity.

Variable	AIC	BIC
family_history_with_overweight	40848.65	40865.40
FAF	40975.70	40992.44
CAEC	41330.75	41364.25
CH20	41521.49	41538.23
FAVC	41560.79	41577.54
MTRANS	41875.93	41917.79
FCVC	42046.29	42063.04

Gender	42289.83	42306.58
SCC	43241.27	42358.02
CALC	42436.82	42470.31

Table 2: Top 10 Most Important Variables Based on AIC and BIC

This approach offered a framework for feature selection for our different methods. However, we acknowledge AIC and BIC are univariate metrics and don't account for interactions or combined effects between variables. Thus when we investigate different methods we will conduct additional feature engineering to enhance the selection process and identify features that might only show significance in combination with others. Overall however, the AIC-BIC evaluation provided a solid starting point for narrowing down the variables and improving model interpretability.

4. Methodology

The Kaggle competition split the dataset into training and testing data. Thus for this section of our project we used the training set given to investigate different classification models that could diagnose obesity for the testing set. We investigated three models: logistic regression, linear discriminant analysis (LDA) with 10-fold CV, and random forest. We then compared these models by evaluating their accuracy.

4.1 Logistic Regression

Logistic regression is used for modeling the relationship between a binary dependent variable and one or more independent variables. This makes it particularly suitable for classification problems, such as predicting whether an individual is obese. One of its strengths is its simplicity and interpretability, as the coefficients provide insights into how each predictor affects the likelihood of the outcome. However, logistic regression assumes a linear relationship between the predictors and the log-odds of the outcome, which may not hold in more complex datasets. Additionally, it may struggle with highly correlated variables or cases with non-linear decision boundaries unless extended with techniques like interaction terms or polynomial features.

For our logistic regression model predicting obesity status, we performed variable selection using forward stepwise selection based on Akaike Information Criterion (AIC) to identify the most significant predictors. The final model included 22 variables, such as family history of overweight, frequency of physical activity (FAF), daily water consumption (CH2O), and dietary habits like vegetable consumption (FCVC) and fast food availability (FAVC). Other

predictors included demographic factors like age, height, and gender, as well as lifestyle variables like smoking status and mode of transportation (MTRANS). This comprehensive set of predictors was designed to capture the multifaceted nature of obesity risk.

The model achieved an accuracy of 75.86% on the training data, which is a promising result, though further validation on unseen data would be necessary to confirm its generalizability. On Kaggle, this model achieved a score of 74.55%. The inclusion of diverse predictors highlights the interplay of behavioral, environmental, and demographic factors in obesity, demonstrating logistic regression's ability to integrate multiple variable types into a unified predictive framework.

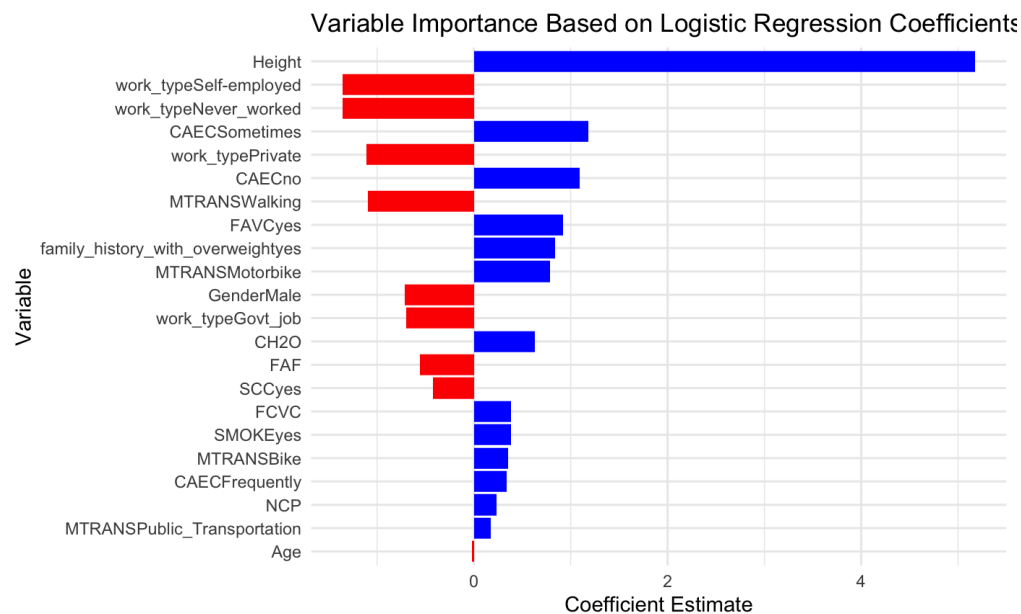


Figure 6: Variable Importance Plot: This plot shows the 22 variables that were selected through forward stepwise AIC. We can see that the most significant variables include height, work type, and walking habits.

Prediction	Not Obese	Obese
Not Obese	1511	443
Obese	265	714

Table 3: Logistic Regression Confusion Matrix: From this confusion matrix, we can see that the logistic regression model achieved an accuracy of 75.86% and a misclassification rate of 24.14%.

4.2 Linear Discriminant Analysis - LDA

Linear Discriminant Analysis is a classification algorithm that performs dimensionality reduction while maximizing class separability. In this project, we used LDA with 10-fold cross-validation to predict obesity based on the following selected features: "Cholesterol", "CH2O", "FCVC", "family_history_with_overweight", "CAEC", and "FAVC". The 10-fold cross-validation was used to evaluate the models performance and ensure stronger results. It essentially re-split the training dataset into 10 equal parts, training the model on 9 folds and testing on the remaining fold. This process was repeated 10 times, with each fold serving as the test set once, reducing the risk of overfitting and increasing the model's generalization power. The cross-validated accuracy provides a more reliable estimate of the LDA model's performance across different subsets of the data.

Before beginning the modeling process, we decided to enhance our feature selection process by running ANOVA on the variables and choosing the best three numerical and categorical variables for our model. For the numerical variables chosen we plotted the following density plots for obese and non-obese subjects to ensure that the distributions were different and the predictors were significant:

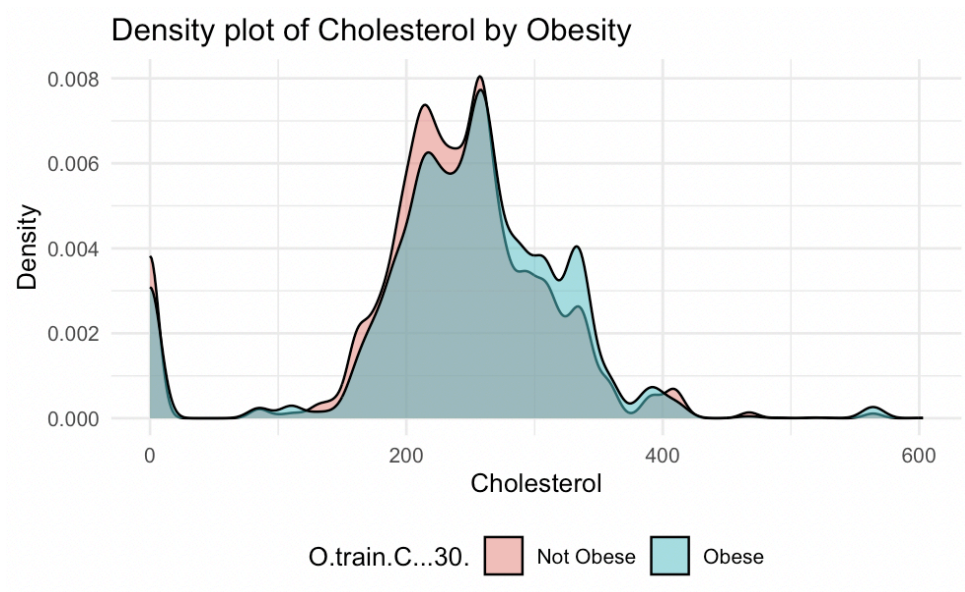


Figure 7: Density Plot of Cholesterol by Obesity. The graph shows a difference between the two density curves, with the “obese” group exhibiting a higher density across the 200-300 cholesterol range and the “not obese” group peaking around lower cholesterol values. This suggests that obesity might be associated with higher cholesterol levels and that cholesterol could be a significant feature for our model.

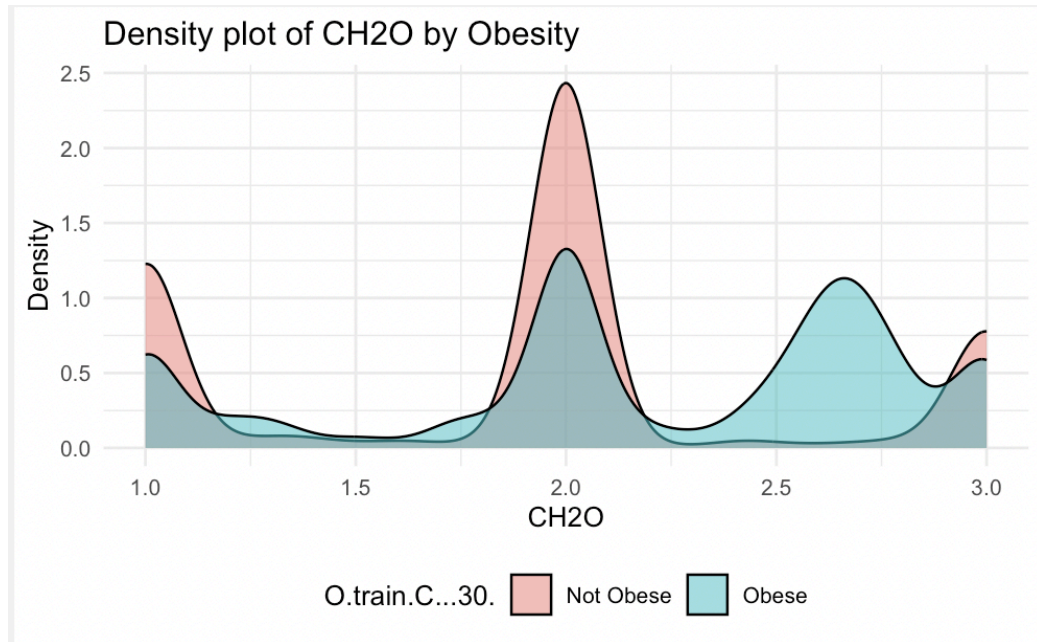


Figure 8: Density Plot of CH2O by Obesity. The graph shows a significant difference between the two density curves, with the “obese” group peaking at CH2O = 2 and CH2O = 2.75 and the “not obese” group showing a dominant peak at CH2O = 2. In addition, the “obese” group shows a broader spread, whereas the “not obese” group concentrated in a more narrow range (1.75-2.25). This suggests that obesity might be associated with more variable or higher CH2O and that CH2O could be a significant feature for our model.

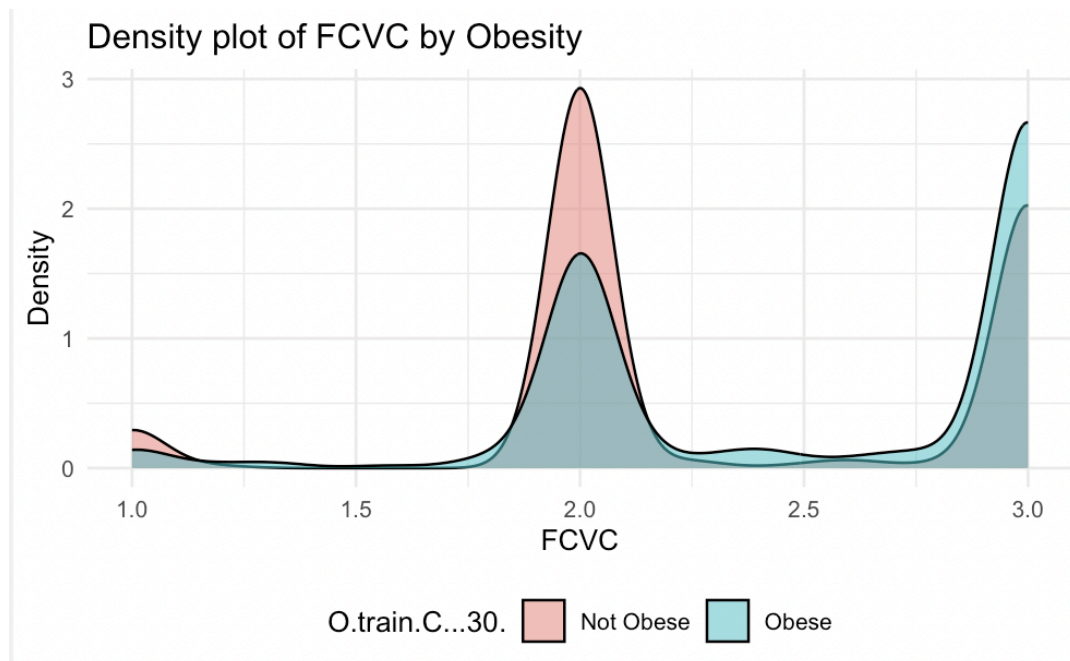


Figure 9: Density Plot of FCVC by Obesity. The graph shows a significant difference between the two density curves, with the “obese” group peaking at FCVC=3 and the “not obese” group peaking at FCVC = 2. While there is some overlap between the groups the distributions diverge for higher values. This suggests that obesity might be associated with higher FCVC and that FCVC could be a significant feature for our model.

Next, for the categorical variables chosen we plotted the following bar plots for obese and non-obese subjects to ensure that the proportions were different and the predictors were significant:

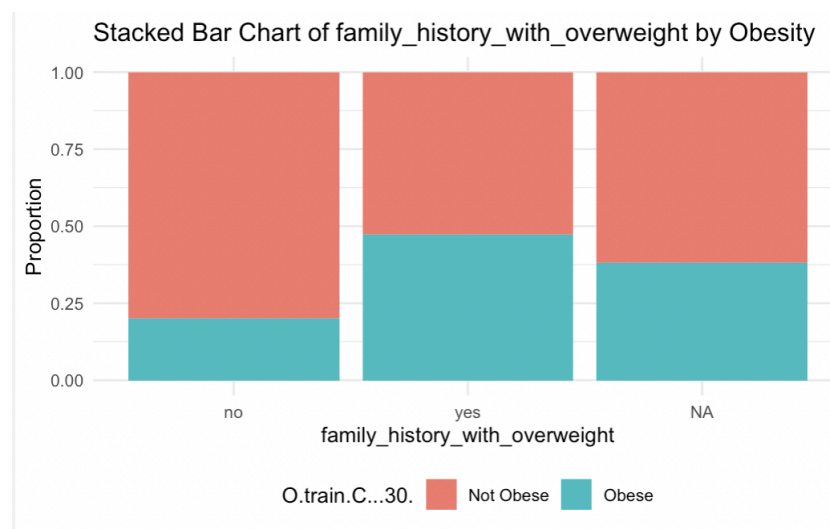


Figure 10: Bar Chart of Family History With Overweight by Obesity. The chart suggests that having a family history of being overweight could be associated with a higher likelihood of obesity, while the absence of a family history appears more common among non-obese individuals.

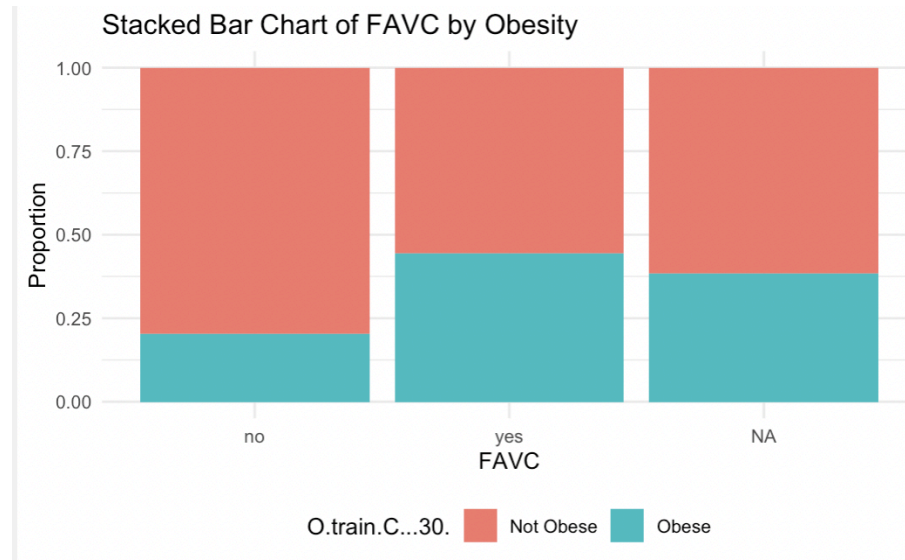


Figure 11: Bar Chart of FAVC by Obesity. The chart suggests that having no FAVC might reduce the likelihood of obesity, while having FAVC is associated with a higher proportion of obesity.

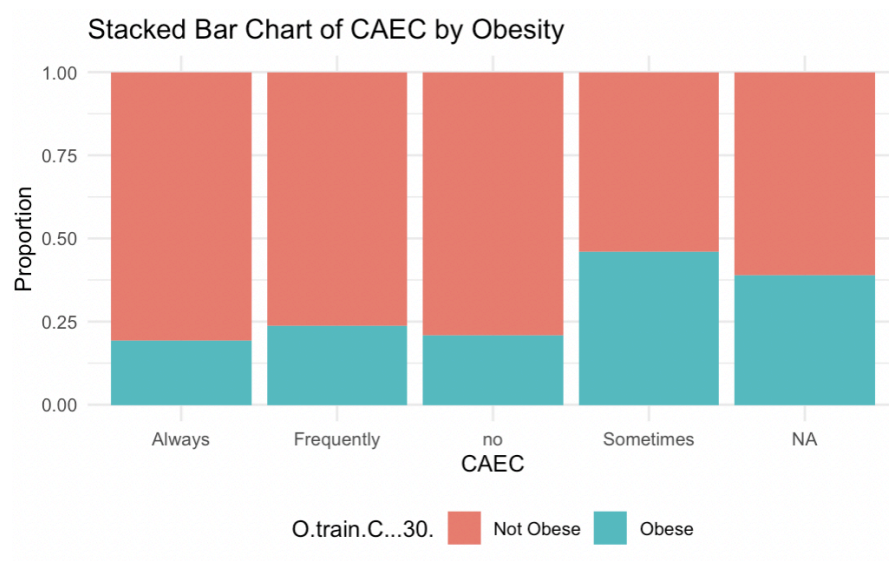


Figure 12: Bar Chart of CAEC by Obesity. There is a clear relationship where higher frequency of CAEC ("Always" and "Frequently") correlates with a higher proportion of obesity, while lower or no CAEC correlates with being "Not Obese."

After identifying our variables, we created our model. This method resulted in an accuracy of 0.7287 and a kappa statistic of 0.4198 on the validation set, which indicates moderate agreement between the predictions and actual labels. The accuracy based on Kaggle score was 0.7112 and the model ranked 17th when it was submitted.

Prediction	Not Obese	Obese
Not Obese	9596	2959
Obese	2325	4597

Table 4: The Confusion Matrix for LDA Model. The confusion matrix above shows a sensitivity of 0.8050 and a specificity of 0.6084, which indicates that the model is better at correctly identifying non-obese individuals compared to obese ones.

While the results of the LDA are promising, one area of improvement is enhancing the model's ability to correctly classify obese individuals. Thus we continued to explore different methods to model our classification problem.

4.3 Random Forest

Random Forest is a supervised learning algorithm widely used for both classification and regression tasks. It operates by constructing multiple decision trees during training and aggregating their predictions to produce a final result, typically based on majority voting or averaging. The process begins with training datasets, from which multiple trees are generated and analyzed to determine a consensus prediction. This model assumes that feature variables contain actual values and that predictions from individual trees have low correlation, enhancing overall accuracy. Random Forest also handles missing values effectively and examines proximity between data points for clustering or outlier detection. Due to its robust structure and ability to reduce overfitting, Random Forest often achieves higher accuracy compared to other models.

We chose the Random Forest model for its accuracy and efficiency. To refine it, we identified the top ten predictors, including height, race, age, physical activity frequency (FAF), caloric intake (CALC), and the number of meals per day (NCP), using feature importance metrics and visual analysis.

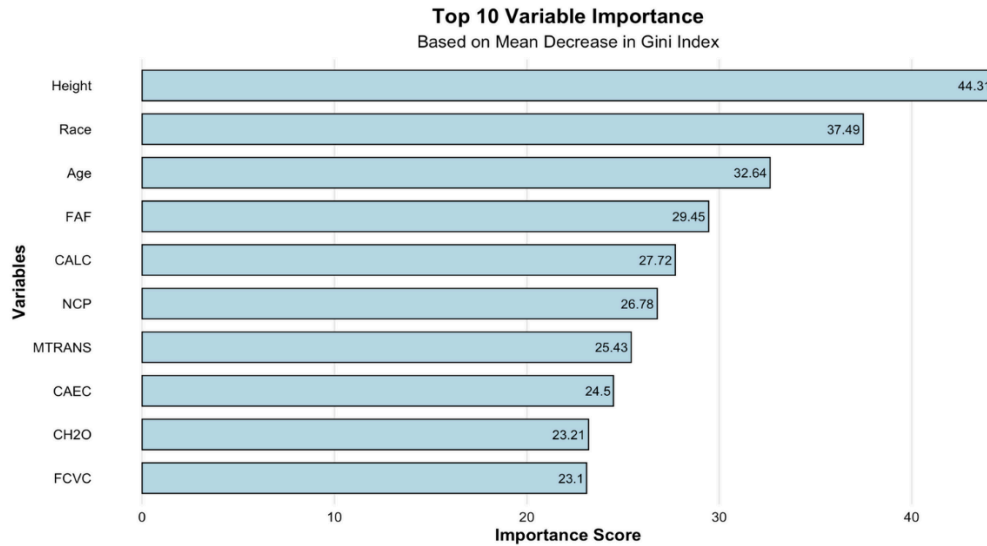


Figure 13: Variable Importance Plot by Random Forest Model. This bar plot illustrates the top 10 predictors ranked by their importance, measured by the mean decrease in Gini index. Height, race, and age emerged as the most influential variables in the Random Forest model.

The model was optimized through hyperparameter tuning by adjusting the number of trees (ntree), variables per split (mtry), and minimum node size (nodesize). Testing various combinations allowed us to balance accuracy and complexity effectively. Validation metrics were then analyzed to minimize misclassification and ensure reliability.

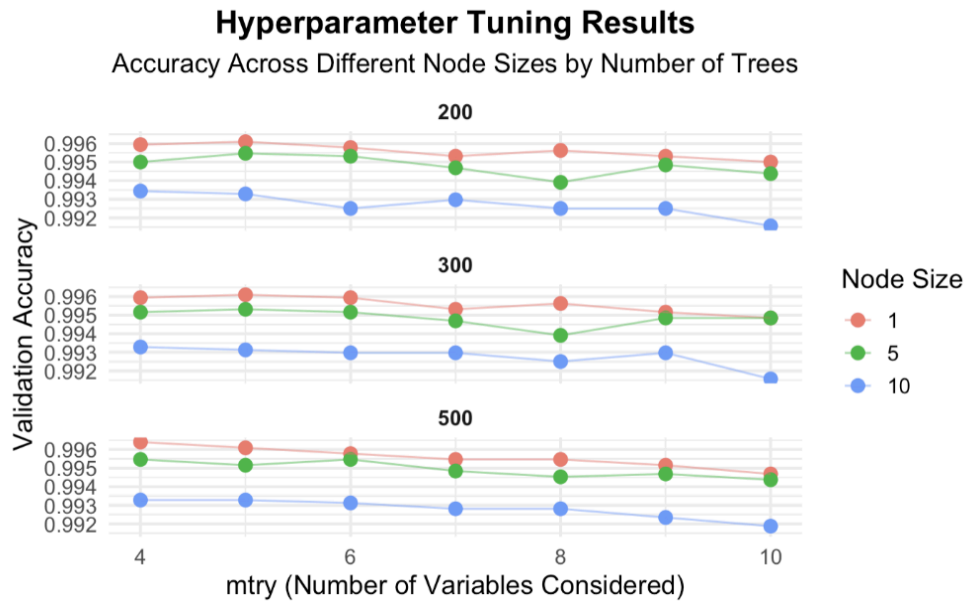


Figure 14: Hyperparameter Tuning Results for the Random Forest Model. These line plots depict the impact of varying the number of trees and node sizes on validation accuracy. The results indicate that the model performs consistently well across different parameter combinations, with slight variation for larger node sizes.

Prediction	Not Obese	Obese
Not Obese	15615	10
Obese	104	9883

Table 5: The Confusion Matrix for Random Forest Model. This table presents the classification results of the Random Forest model, showing the predicted versus actual categories for "Obese" and "Not Obese." The model correctly classified 15,615 "Not Obese" instances and 9,883 "Obese" instances, with minimal misclassification errors (10 false positives and 104 false negatives).

The final model achieved a classification accuracy of 99.78% and a misclassification rate of 0.0081. This demonstrates its high performance, achieving high accuracy with minimal misclassification errors, as demonstrated by our classification matrix. Its ability to handle complex data relationships and prioritize certain predictors made it the best choice for this analysis, ensuring both reliability and precision in the results.

5. Limitations and Discussions

Our best model (Random Forest) achieved a testing accuracy of 99.7% and got 15th on the leaderboard. Although the accuracy is promising, we recognize the limitations of our methods and our modeling process overall.

Starting with our modeling processes, the challenges we faced with increasing accuracy can be attributed to our data cleaning process. First, we chose to deal with numerical missing values by imputing them with the median, however this assumes a normal distribution that may not exist in all variables. To improve the model, we could try imputing using the K-Nearest Neighbors method. Second, the models' performances could potentially be affected by multicollinearity among features. Furthermore, we assumed additivity and relevance of the features, which may not be true. While we tried to choose only relevant variables by looking at AIC, BIC, and through forward selection, we could have tried more methods to verify relevancy of the features. Lastly, we didn't have much context on the study and how the data was collected, thus our results may have been affected because we assumed independence between observations. In our next steps we would conduct further research on the study and on the variables that could deepen our understanding of the issue and help us fine-tune our model.

In terms of limitations of the modeling methods we attempted, first Logistic Regression is a very simple and interpretable model that doesn't have any distribution assumptions, however it is limited by linearity. If the decision boundary isn't linear and the relationship between the predictors and obesity status is more complex a more flexible model would be more suitable.

Next, LDA is a strong contender for a model because it includes dimensionality reduction and accounts for feature importance in the modeling process. However, it is a less flexible model because it assumes that the predictors follow a Gaussian distribution and that all classes have equal covariance matrices. If these assumptions don't hold for our dataset, the model's effectiveness will be reduced.

Lastly, Random Forest is very flexible and can handle more complex relationships between predictors. However, since it is more complex, it is less interpretable. Additionally, its predictions are very sensitive to parameter tuning, making it more unstable. Lastly, it is a more computationally expensive method than the other methods.

From the confusion matrices, we observed that our models generally performed better at classifying non-obese individuals than obese ones. In the real-world, typically accurate identification of obese individuals is more crucial for timely interventions and public health planning. In our next steps, we could benefit from addressing these limitations by experimenting with more non-linear models and implementing more preprocessing techniques.

6. Results and Conclusions

Overall, our model achieved a high accuracy of 0.997 in predicting obesity status, demonstrating the effectiveness of the Random Forest model in this context. However, we acknowledge some limitations in our approach that may have impacted the model's performance. For example, during data cleaning, missing values were imputed using median and mode imputation, which assumes that the underlying data distributions are normal. While this method allowed us to handle missing data efficiently, it may have introduced biases or affected the predictive power of certain variables. Additionally, our assumptions about how the observations were gathered and the relationships between predictors could limit the model's generalizability to different populations or datasets.

In the context of the Kaggle competition, while our model had almost perfect testing accuracy, further improvements could be made to increase our position on the leaderboard. For example, we could incorporate more advanced imputation techniques, such as multiple imputation or KNN imputation. For our next steps, we aim to dive deeper into the dataset and variables to refine our understanding of obesity-related factors and the study itself. Further research into the relationships between predictors could provide a more holistic view of the issue. These efforts could help us fine-tune our approach and improve our model's performance even more. Overall, this project was extremely beneficial in helping us improve our modeling and data analysis skills.

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